



The Role of Generative AI in Supporting Neurodiverse Individuals: Literature Insights and Future Directions

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Abstract. Neurodiversity emphasizes the natural variation in human cognition, celebrating the unique strengths and perspectives that neurodiverse individuals bring to society. This paper explores the role of artificial intelligence (AI) in supporting neurodiverse individuals, focusing on both traditional and generative AI systems. Current trends reveal two primary applications: (1) detecting, diagnosing, and treating neurodiverse conditions and (2) enhancing cognitive abilities and promoting wellbeing. The shift from a medical model to a social and ecological model underscores the need for inclusive and strengths-based AI systems that empower neurodiverse individuals across various domains, such as education, employment, and social interaction. By identifying design requirements—including customizability, ethical considerations, and responsible design practices—this paper highlights how generative AI can address current gaps and expand opportunities for neurodiverse populations. Future research should prioritize participatory design, inclusive datasets, and applications that holistically support neurodiverse individuals in achieving their potential.

Keywords: Neurodiversity · Generative AI · Inclusive Design

1 Introduction

There are over 13 million adults in the United States with an autism spectrum disorder (ASD) or attention-deficit/hyperactivity disorder (ADHD); this includes approximately 1-2 Neurodiversity refers to the natural variation in human cognition and behavior. Rather than viewing neurological differences as deficits, the concept of neurodiversity emphasizes the unique strengths and perspectives that individuals with different neurological wiring bring to society. Neurotypical, on the other hand, describes individuals whose neurological development and functioning fall within the dominant societal standards [6]. Some common neurodiverse conditions include attention-deficit/hyperactivity disorder (ADHD), autism spectrum disorder (ASD), dyslexia, down syndrome, and cerebral palsy. While neurodiverse individuals face certain challenges in conventional social and learning environments, they also possess unique strengths, such

as, hyperfocus, creativity, pattern recognition, visual-spatial skills, attention to detail, and innovative thinking [14]. There is a growing movement to recognize neurodiversity not as a series of disabilities, but as a spectrum of cognitive differences. By embracing neurodiversity and tapping into the resource pool of neurodivergent workforce, organizations can leverage the strengths and perspectives of these employees to improve creativity and performance of the organization. This shift necessitates understanding and accommodation rather than treatment and normalization. Recognizing and valuing neurodiversity is not just an ethical imperative, but a strategic advantage. By embracing the unique cognitive processes of neurodivergent individuals, society can benefit from their contributions in various fields, including education, technology, and the workplace [11].

Although the term neurodiversity encompasses a range of cognitive and intellectual disabilities, much of the research in information and computational sciences focuses heavily on Autism Spectrum Disorder (ASD) and, to a lesser extent, Attention-Deficit/Hyperactivity Disorder (ADHD). Autism Spectrum Disorder is the most frequently discussed neurodiverse condition. Many studies examine the characteristics of ASD, prevalence rates, and strategies for supporting individuals with ASD, including the application of AI [2, 12, 16, 18, 19, 22]. ADHD is the second most frequently explored neurodiverse condition in this context [9, 11, 17]. Some studies address the challenges faced by individuals with ADHD, while others investigate AI-based interventions to support them. Dyslexia is also mentioned in several studies, which discuss it as a neuro-diverse condition with unique strengths and challenges. These studies explore potential AI applications designed to assist individuals with dyslexia.

Other neurodiverse conditions, such as Tourette's Syndrome, Down Syndrome, Cerebral Palsy, Epilepsy, Bipolar Disorder, Dyspraxia, Depression, and Alzheimer's Disease, are either briefly mentioned or absent in the recent literature reviewed. However, when specifically searching for these disabilities, some research does emerge, though much of it adopts a medical model perspective. This means the focus is often on detection, diagnosis, and treatment rather than broader applications, such as leveraging AI for support [4, 13, 20].

Advances in generative AI provide a tremendous opportunity to support learning of neurodiverse students who often struggle to keep up with other students in the class. The goal is not to separate these students but to equip them, so they are able to compensate for their shortcomings and are able to keep up with the class curriculum. Neurodiverse students can leverage AI tools such as text-to-speech for efficient notetaking, those with speech and communication difficulties can utilize AI tools to express themselves effectively and participate in classroom discussions when they wouldn't have been able to in the past. AI can also be used to create personalized curricular content catering to different learning needs for instance text-to-speech and speech recognition tools can break down barriers for students with dyslexia or auditory processing issues. Similarly, lesson plans can incorporate visual aids like charts, diagrams, icons, and color coding to present information in a clear, structured way, breaking down complex concepts into manageable chunks, and allowing for easy navigation and

comprehension, for neurodiverse students who often struggle with traditional text-based learning methods.

There remains a significant gap in AI and technology research for these conditions. This disparity may stem from differences in the diagnostic processes and inherent nature of these disabilities compared to ASD and ADHD. Addressing this gap represents an important opportunity for future research. The current research on AI design for neurodiversity can be broadly categorized into two main areas: (1) the use of AI to detect, diagnose, and treat neurodiverse individuals, and (2) the use of AI to enhance cognitive abilities and promote the wellbeing of neurodiverse individuals in various social contexts.

Several studies focus on leveraging AI to identify and diagnose neurodiversity, particularly in conditions such as Autism Spectrum Disorder (ASD) and Attention Deficit Hyperactivity Disorder (ADHD). AI and deep learning techniques are advancing the detection and diagnosis of neurodiverse conditions, particularly Autism Spectrum Disorder (ASD) and Attention Deficit Hyperactivity Disorder (ADHD). For ASD, [16] demonstrated how AI can analyze behavioral features from video data to detect the condition, offering a faster diagnostic process without compromising accuracy. Researchers have also explored AI-powered tools like Robot Intelligent Modules (RIM) and Behavior Manager (BM) to assist in autism treatment. [1] highlight the use of federated learning, combined with classifiers such as logistic regression and support vector machines, to analyze data from children and adults for detecting autistic traits. Deep learning techniques, including facial image analysis, are being applied for ASD detection, with studies using computer vision to estimate engagement intensity in children with ASD. For ADHD, [9] explored the use of deep learning, particularly Convolutional Neural Networks (CNNs), to improve the accuracy and reliability of ADHD screening and diagnosis in children. Their work reviewed various deep learning models, such as those analyzing EEG data and employing computer vision techniques, showcasing the ability of these technologies to process large datasets and extract implicit patterns critical for diagnosis.

The second major area of research focuses on utilizing AI to support and enhance the cognitive abilities and wellbeing of neurodiverse individuals. Research has increasingly focused on leveraging AI and emerging technologies to enhance the cognitive abilities, communication skills, and social interactions of neurodiverse individuals. The Neurodiversity Reciprocating Wellbeing Artificial Intelligence (NRWAI) framework, proposed by [6], aims to enhance cognitive functioning and wellbeing through personalized AI-driven learning experiences and support systems tailored to the unique needs of neurodiverse individuals. Additionally, [22] highlight the potential of AI to reduce cognitive load during social interactions, addressing a key challenge for many neurodiverse individuals. AI is also being used to create virtual environments where neurodiverse individuals can practice essential social skills, such as communication and interaction, as discussed by [19]. An example is the “emotion translator” prototype described by [22], which helps users understand and express emotions during conversations. These advancements illustrate the broader applications of AI in

enhancing communication skills and self-regulation for neurodiverse populations. Emerging technologies such as voice assistants, wearables, and virtual reality further contribute to supporting neurodiverse users in areas like executive functioning, communication, and interpersonal skills. [14] emphasize the importance of involving neurodiverse users in the design process to create inclusive and effective solutions.

Our focus in this research is on the second area of research i.e., support and enhance cognitive abilities and well-being of neurodiverse individuals especially towards the goal of integrating them in society seamlessly by compensating for any lack of abilities. The rest of the paper discusses the extant literature on the topic as well as future directions of research.

2 Shift from Medical Model to Social Wellbeing

The two topics—AI for Detecting, Treating, and Diagnosing Neurodiverse Individuals and AI for Enhancing Cognitive Abilities and Wellbeing—can be discussed in light of [8], which contrasts the traditional medical model of disability with the social and ecological model. The medical model views disability as a pathology with the goal of curing or normalizing individuals to resemble typically developing peers. AI research aligned with this model often focuses on detecting autism through behavioral and biological markers tied to diagnostic criteria. These models emphasize identifying and categorizing atypical traits. However, the medical model perspective often fails to recognize the societal barriers that contribute to disability and can overlook the strengths and unique perspectives of neurodivergent individuals. This has led to a growing shift in research in recognizing that disability is a complex phenomenon arising from the interaction between an individual's characteristics and the environment in which they live. Hence, interventions should be tailored to individual needs and preferences, addressing both individual challenges and environmental barriers. This perspective promotes the development of AI-driven solutions that are inclusive, celebrate neurodiverse strengths, and create environments where individuals can thrive [10,11]. This shift in focus has led to innovations in AI design and development to support neurodiverse individuals in a variety of contexts such as education, interpersonal communication, and social wellbeing.

2.1 AI for Cognitive Enhancement and Wellbeing

AI-driven technologies have significantly improved cognitive accessibility for neurodiverse individuals, serving as gateways to new modes of interaction and understanding. These tools enhance cognitive abilities and promote well-being by providing personalized learning experiences tailored to individual needs and learning styles. AI-powered assistive technologies also play a crucial role in supporting various aspects of daily living, including sensory processing and emotional regulation. By offering individualized support, these tools enhance the quality of life and overall well-being of individuals with cognitive disabilities [7].

The adaptability of AI enables personalized learning, adaptive feedback, and context-aware interactions, which address the unique challenges faced by neurodiverse learners. Such capabilities can lead to improved learning outcomes and a deeper sense of well-being, helping individuals thrive in educational and everyday contexts [6]. Through its ability to tailor support to diverse needs, AI continues to transform opportunities for cognitive accessibility and empowerment among neurodiverse populations.

2.2 AI for Communication and Social Skills

AI has significant potential to create environments that foster communication and social interaction among neurodiverse individuals. For example, [15] highlights the effectiveness of the ECHOES project, which used AI-powered tools to motivate social interaction and communication in autistic children who might otherwise struggle with these skills. The project underscores the importance of blending AI and human support to create a comprehensive and adaptable learning environment. The presence of a human practitioner provided individualized support and intervention, addressing each child's specific needs and enhancing the overall effectiveness of the AI tools.

Similarly, conversational agents have been employed to help children with Autism Spectrum Disorder (ASD) develop emotional and social skills [18]. These agents act as personal assistants, narrate social stories, and facilitate communication. Generative AI, which leverages deep learning to create content, shows great promise in personalizing and adapting emotional learning experiences, particularly for children with high-functioning autism. Beyond serving as personal assistants, AI tools for communication and social support must move beyond mere detection and labeling of emotions. Instead, they should facilitate the co-construction of emotional understanding between autistic and neurotypical conversation partners.

For instance, [22] explored the use of AI-driven virtual environments where neurodiverse individuals could practice essential social skills like communication and interaction. One innovation was the development of an “emotion translator,” designed to encourage active interpretation, negotiation, and shared meaning making regarding emotions. This approach fosters a more collaborative and dynamic communication process, emphasizing mutual understanding rather than one-sided adaptation. By integrating these tools into interactive environments, AI can play a transformative role in enhancing social and emotional learning for neurodiverse individuals.

2.3 AI for Education and Employment

AI has significant potential to advance education and employment opportunities for neurodiverse individuals by addressing their unique challenges and leveraging their strengths. In education, wearable technology, such as smartwatches, can monitor student activity and provide real-time interventions to support learning. These devices can collect sensor data to train activity recognition models

capable of identifying and differentiating between activities like reading, writing, or being off-task. For example, if the smartwatch detects prolonged inactivity, it could issue a gentle reminder to reengage with the task, helping neurodiverse students maintain focus and improve their learning outcomes. The development of such tools requires inclusivity and careful consideration of diverse student populations, ensuring that AI systems are free from biases and designed to meet a wide range of needs [21].

In the realm of employment, AI-powered tools can address barriers faced by neurodiverse individuals in job searching and workplace integration. These tools can assist with tasks such as resume writing, interview preparation, and skill development, while also fostering more inclusive work environments. For instance, AI-driven job matching systems can pair neurodiverse job seekers with employers who value their specific skills and are prepared to offer necessary accommodations. Similarly, AI can analyze job descriptions to identify key skills and requirements, tailoring resumes and cover letters to better align with specific opportunities. Virtual interview coaches provide another avenue of support, offering feedback on communication style, eye contact, and body language to help neurodiverse individuals build confidence. AI can also support neurodiverse employees in managing their emotions and sensory sensitivities, which are often critical challenges in workplace settings. Tools designed to identify triggers and develop coping strategies can help individuals navigate sensory overload and social interactions effectively [2, 5].

3 Design Requirements for AI for Neurodiversity

The existing literature identifies several critical design requirements for AI systems aimed at supporting neurodiverse individuals. These requirements emphasize the need for personalization, inclusivity, and ethical considerations to ensure technology effectively meets the unique needs of this population. These design requirements have been summarized in Table 1.

3.1 Customizability: Tailoring AI Systems to Individual Needs

Customizability is a cornerstone of effective AI design for neurodiverse populations, recognizing that a one-size-fits-all approach is inherently inadequate. AI-powered tools for learning and communication can offer personalized experiences that adapt to individual cognitive patterns, sensory sensitivities, and learning styles. For example, AI applications designed for learners with ADHD can adjust the pace and delivery of educational content based on the learner's attention span and engagement level. These tools might provide multiple modalities, such as visual or auditory options, to align with individual preferences and enhance comprehension [7].

In social communication, AI-driven technologies have shown promise in creating tailored interventions. The ECHOES project, for instance, utilized an AI-powered virtual agent named Andy to support neurodiverse children in developing social and communication skills. By adapting to the child's actions and

Table 1. Design Requirements for AI in Neurodiversity

Category	Requirement
Customizability	Personalized tools that adapt to cognitive patterns, learning styles, and sensory preferences. Customizable interfaces for sensory needs, such as brightness and sound adjustments. Adaptive social tools, like emotion translators and tailored learning experiences.
Strengths-Based Approaches	Emphasize individual strengths in education and employment. Support job-seeking with collaborative tools and job simulations.
Ethical Considerations	Train AI on diverse datasets to mitigate biases. Ensure equitable access and culturally inclusive design. Maintain transparency in decision-making processes.

providing individualized support, Andy created a more engaging and effective learning environment. Similarly, EmoEden, an AI tool powered by large language models (LLMs), personalizes emotional learning for children with High Functioning Autism (HFA) by generating tailored visual and textual content. This individualized approach helps children better understand emotions in ways that resonate with their abilities and preferences [15, 18].

Customizability also extends to tools that bridge emotional understanding. For example, a prototype featuring a customizable matrix of emotion pictograms allowed autistic adults to create personalized emotion taxonomies. These tools help address challenges in interpreting and expressing emotions, fostering more meaningful social interactions [22].

3.2 Strengths-Based Approaches: Focusing on Abilities, Not Deficits

A strengths-based approach shifts the focus from compensating for perceived deficits to leveraging the unique abilities of neurodiverse individuals. In education, AI-powered tools can identify and highlight students’ strengths, offering personalized learning plans that emphasize areas of excellence. For instance, if a learner exhibits exceptional problem-solving skills, the AI could design tasks that build on this ability, fostering confidence and motivation.

In employment, AI can similarly support strengths-based practices by helping employers identify and capitalize on neurodiverse candidates’ unique talents. For example, AI systems can analyze resumes and work histories to detect patterns of success, helping match candidates with roles that suit their abilities. Job simulations powered by AI can allow candidates to demonstrate their skills in real-world contexts, offering both employers and candidates a clearer understanding of fit [11].

Collaborative job-seeking platforms present another opportunity for strength-based approaches. These systems can facilitate communication and

preparation by enabling neurodiverse individuals to collaborate with family members, mentors, and experts. Features such as socially supported communication tools can help users refine emails to potential employers, ensuring appropriate tone and structure. Similarly, mock interview simulations with AI can provide constructive feedback, helping neurodiverse individuals develop their skills in a supportive environment [2].

3.3 Ethical Considerations: Ensuring Fairness, Inclusivity, and Transparency

Ethical considerations are paramount in designing AI systems for neurodiverse individuals, given the potential for algorithmic bias and unequal access to technology. AI algorithms often learn from datasets that may not fully represent the diverse range of cognitive abilities and experiences, risking the perpetuation of existing biases. For example, an AI tool developed to assist dyslexic students might be less effective if trained predominantly on data from neurotypical learners, potentially misinterpreting their needs or providing suboptimal support [7]. To mitigate such risks, diversity in training data is critical. Including inputs that reflect the lived experiences of neurodiverse individuals ensures that AI systems can address a broader spectrum of needs. Counterfactual data augmentation, which involves simulating alternative viewpoints and scenarios, can help AI systems develop a more nuanced understanding of neurodiverse conditions, reducing oversimplifications and biases [11].

Equitable access to AI-powered tools is another essential ethical consideration. Without targeted efforts, the digital divide could deepen, leaving socioeconomically disadvantaged neurodiverse learners behind. Addressing this requires not only the provision of technology but also robust training for educators and comprehensive support systems for learners. Designing culturally sensitive and inclusive tools is equally important to ensure that all students, regardless of background, can benefit from AI-enhanced learning.

Transparency is also crucial for building trust and fostering informed use of AI tools. Neurodiverse users, as well as educators and employers, should have a clear understanding of how AI systems operate, their limitations, and any potential biases. Educating users about AI interactions prepares them to navigate the digital world more effectively and empowers them as informed participants [7].

3.4 Responsible and Inclusive Design Practices

Responsible design and implementation are essential to ensuring that AI systems for neurodiversity truly meet the needs of their users. User-centered design (UCD) principles advocate for the active involvement of neurodiverse individuals throughout the design process, from conceptualization to evaluation. Frameworks like the Diversity for Design (D4D) approach encourage participatory methods that adapt to users' needs, leveraging strengths while minimizing potential difficulties [3].

Sensitivity to neurodiversity in design requires continuous learning and adaptation. Developers must prioritize creating interfaces that accommodate sensory sensitivities, such as reducing visual clutter or offering multiple interaction modes (e.g., text and voice). Additionally, AI tools should complement, rather than replace, human expertise, particularly in areas like diagnostics, where the nuanced judgment of a professional is indispensable [15].

4 Future Research Opportunities Identified for Generative AI in Neurodiversity

Generative AI presents unique opportunities to transform how technology supports neurodiverse individuals. Unlike traditional AI systems, generative AI can adapt dynamically to individual needs, addressing gaps in therapeutic, educational, and daily living support.

One promising area is **personalized learning**. Generative AI can create tailored educational tools that adapt to individual cognitive strengths and weaknesses. Frameworks like the Neurodiversity Reciprocating Wellbeing Artificial Intelligence (NRWAI) demonstrate how personalized feedback and rewards can enhance engagement and learning outcomes [6]. Similarly, **adaptive communication** tools, such as emotion translators can help neurodiverse individuals navigate social interactions by facilitating mutual understanding [22]. We need to be able to create customized learning plans for neurodiverse students that best match the cognitive abilities of neurodiverse individuals. Currently, the lesson plans for such individuals are done manually that can overwhelm caregivers and teachers. Generative AI can help create initial custom learning plans for individuals more efficiently, providing support to caregivers and educators.

Another area where AI technologies can enhance the quality of life on neurodiverse individuals is by providing them compensating technologies for their disability. For instance, providing real speech to text convertors for hearing impaired, visual to audio summarization for visually impaired, and text to speech conversion for speech impaired individuals. Using these tools in real-time can enable such individuals to normalize themselves in the society and become more self-reliant. This will reduce both the burden of caregivers as well as contributing to societal productivity through participation in jobs requiring higher levels of cognitive skills.

Generative AI also offers potential for **just-in-time therapeutic interventions**, addressing challenges that traditional therapies cannot. Smartwatches and other wear-able devices powered by AI can provide real-time emotional and behavioral support, helping individuals manage transitions and navigate complex tasks [18]. Additionally, AI-driven job-matching systems could revolutionize employment opportunities by aligning neurodiverse skills with workplace needs, promoting inclusivity and reducing barriers. Generative AI enables a shift from deficit-based models to **strengths-based paradigms**. By focusing on unique neurodiverse abilities, generative AI can challenge stereotypes, foster inclusivity, and empower individuals to thrive in diverse social contexts [11]. These

opportunities highlight the transformative potential of generative AI to enhance neurodiverse lives.

Designing generative AI for neurodiverse individuals requires a holistic approach that embraces diversity and prioritizes inclusion. This entails moving beyond existing adaptations to create systems that recognize neurodiversity as a valuable form of human variation. A critical consideration is the **prioritization of participatory design**, where neurodivergent individuals are actively involved throughout the development process [2]. Shifting from “design for” to “design with” ensures that systems reflect lived experiences and diverse perspectives. Overcoming scalability challenges in participatory design requires innovative recruitment strategies, combining online and offline methods, and leveraging advanced tools like natural language processing to analyze large datasets. Equally important is the creation of inclusive datasets. These datasets should capture the full spectrum of neurodiverse experiences, ensuring representation across demographics, disabilities, and cultural contexts [22]. Open-sourcing anonymized data can accelerate collaboration and reduce biases in AI systems. Generative AI must also account for **intersectionality**, addressing the unique challenges faced by individuals with multiple marginalized identities. Finally, generative AI should expand into domains beyond education and employment, such as **mental health, financial management, and social activities**. By providing tools for emotional regulation, personalized budgeting, and accessible navigation, generative AI can holistically support neurodiverse individuals. These considerations will ensure that generative AI is not only assistive but also empowering, enabling individuals to lead fulfilling and independent lives.

5 Conclusion

Artificial intelligence has the potential to transform the lives of neurodiverse individuals by addressing their unique challenges and leveraging their strengths. From enhancing cognitive abilities and communication skills to supporting education and employment, AI-driven solutions are shifting from a medical model to a strengths based and socially inclusive paradigm. Generative AI, in particular, offers unprecedented opportunities for personalized, real-time support, enabling neurodiverse individuals to thrive in diverse contexts. To achieve this vision, the design and development of AI must prioritize customizability, inclusivity, and ethical considerations. Incorporating neurodiverse perspectives through participatory design, creating representative datasets, and addressing biases are essential to building systems that reflect the diversity of human experiences. Future research must continue exploring innovative applications of generative AI, expanding its reach into areas like mental health, financial management, and social activities. By fostering a collaborative and inclusive approach, AI can become not just a tool for assistance but a catalyst for empowerment, enabling neurodiverse individuals to contribute meaningfully and achieve independence in all facets of life. This holistic perspective positions AI as a transformative force in realizing the full potential of neurodiversity.

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